Cloud Computing Lab Eval Missed

Project Report : Cloud Article Deployment on Amazon EKS

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Project Overview:

This project demonstrates my practical implementation of containerizing and deploying a web-based news article focused on "Cloud Computing" using highly scalable cloud infrastructure on Amazon Web Services (AWS). I built this to showcase my understanding of modern cloud architectures.

Prerequisites:

1.AWS CLI

2.Docker Desktop

3.kubectl

4.eksctl

Configure AWS Credentials :

1. Log in to your AWS Academy Learner Lab.

2. Click on AWS Details to reveal your temporary credentials.

“aws_access_key_id=YOUR_ACCESS_KEY
aws_secret_access_key=YOUR_SECRET_KEY
aws_session_token=YOUR_SESSION_TOKEN”

Application Containerization:

1. Create the Frontend Code, Create a file named index.html with your web content
2. Create the Dockerfile In the same directory as index.html, create a file exactly named Dockerfile:

“FROM nginx:alpine COPY index.html /usr/share/nginx/html/index.html EXPOSE 80 CMD ["nginx", "-g", "daemon off;"]”

3. Build the Docker Image Run the following command in your terminal to build the image tagged as cloud-news-frontend:

“docker build -t cloud-news-frontend”

Push to Amazon Elastic Container Registry (ECR) :

1. Create an ECR Repository Create a repository in AWS to hold your Docker image:

   “aws ecr create-repository --repository-name cloud-news-frontend --region us-east-1”

2. Authenticate Docker with ECR

   “aws ecr get-login-password --region us-east-1 | docker login --username AWS --password-stdin YOUR_ACCOUNT_ID.dkr.ecr.us-east-1.amazonaws.com”

3. Tag and Push the Image Tag your local image to match the remote ECR repository, then push it:

   “docker tag cloud-news-frontend:latest YOUR_ACCOUNT_ID.dkr.ecr.us-east-1.amazonaws.com/cloud-newsfrontend:latest docker push YOUR_ACCOUNT_ID.dkr.ecr.us-east-1.amazonaws.com/cloud-news-frontend:latest”

Create the Amazon EKS Cluster:

1. You cannot use standard eksctl commands because you lack the permissions to create new IAM roles. You MUST use the pre-existing LabRole. Create a configuration file named cluster.yaml:

   “apiVersion: eksctl.io/v1alpha5 kind: ClusterConfig metadata: name: cloud-project-cluster region: us-east-1 version: "1.30" iam: serviceRoleARN: "arn:aws:iam::YOUR_ACCOUNT_ID:role/LabRole" managedNodeGroups: - name: standard-nodes instanceType: t3.medium minSize: 1 maxSize: 3 desiredCapacity: 2 iam: instanceRoleARN: "arn:aws:iam::YOUR_ACCOUNT_ID:role/LabRole" “

2. Run the following command to build the VPC and the EKS cluster:

   “eksctl create cluster -f cluster.yaml”

Kubernetes Deployment:

1. Create Kubernetes Manifests Create a file named k8s-deploy.yaml. Make sure to replace YOUR_ACCOUNT_ID!

   “apiVersion: apps/v1
   kind: Deployment
   metadata:
   name: cloud-news-frontend
   spec:
   replicas: 2
   selector:
   matchLabels:
   app: cloud-news
   template:
   metadata:
   labels:
   app: cloud-news
   spec:
   containers:
   - name: frontend

image: YOUR_ACCOUNT_ID.dkr.ecr.us-east-1.amazonaws.com/cloud-news-frontend:latest
ports:
- containerPort: 80
---
apiVersion: v1
kind: Service
metadata:
name: cloud-news-service
spec:
type: LoadBalancer
selector:
app: cloud-news
ports:
- protocol: TCP
port: 80
targetPort: 80”

2. Deploy the Details Apply the manifests to your cluster:

   “kubectl apply -f k8s-deploy.yaml”

3. Verify and Access the Application Check if the pods are running:

   “kubectl get pods”

4. Get the public URL (External-IP) of the LoadBalancer:

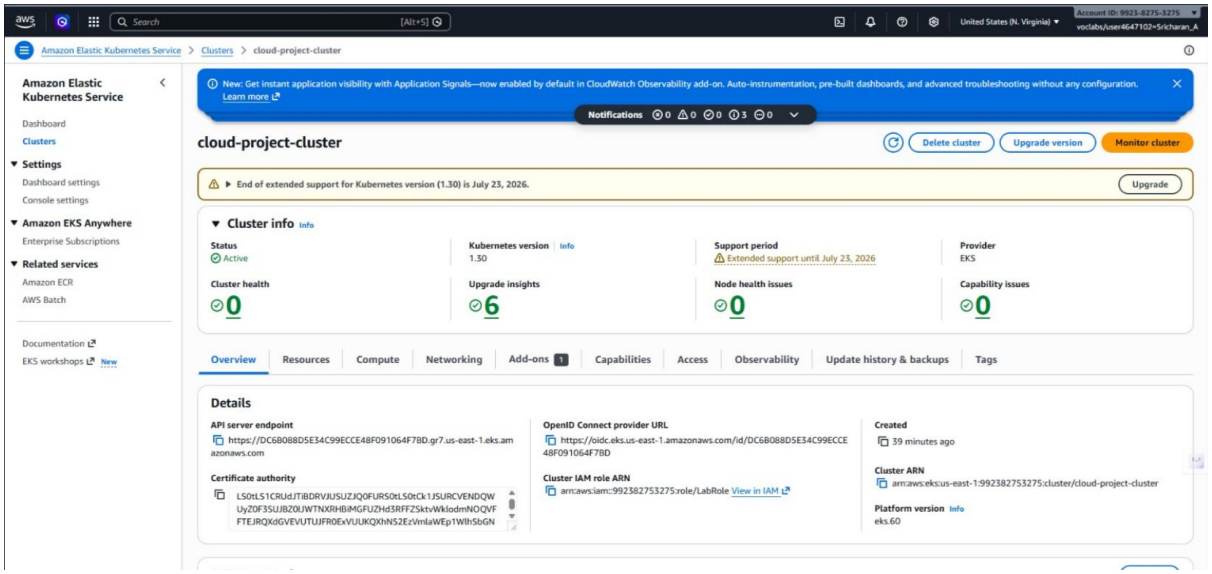
   “kubectl get services cloud-news-service”

AWS Services Used :

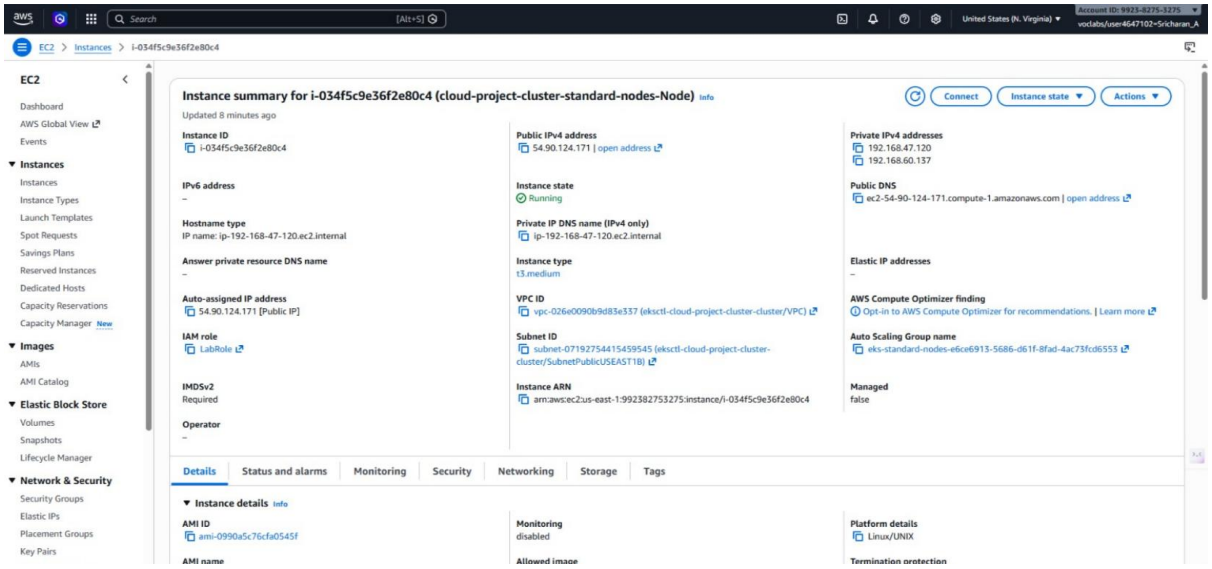
Service	Role in Project
Amazon EKS	Managed Kubernetes service used to orchestrate and run the containerized frontend.
Amazon ECR	Docker container registry used to store and manage the application images.
AWS IAM	Managed access using the `LabRole` to ensure secure interactions between EKS and other AWS services.
Amazon VPC	Networking foundation (Subnets, Route Tables, NAT Gateways) created by `eksctl` to isolate the cluster.
Elastic Load Balancing (ELB)	Automatically provisioned by the Kubernetes Service to provide a public URL for the application.
Amazon EC2	The underlying virtual machines (t3.medium) that acted as worker nodes in the EKS cluster.
CloudFormation	Used by `eksctl` as the "Infrastructure as Code" engine to provision the entire stack.

Project Screenshots :

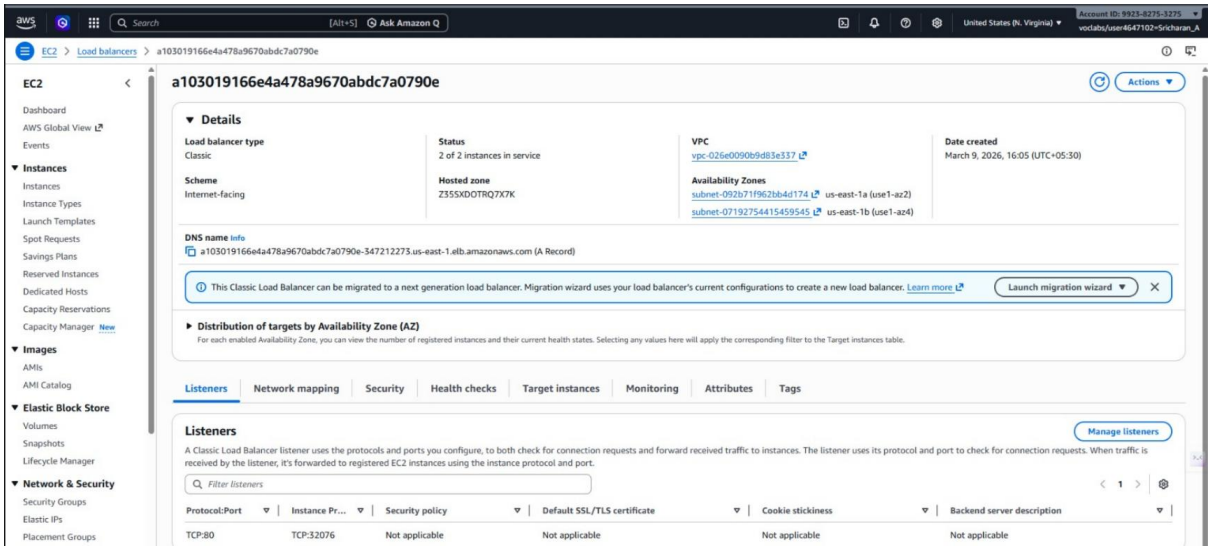
Cluster :



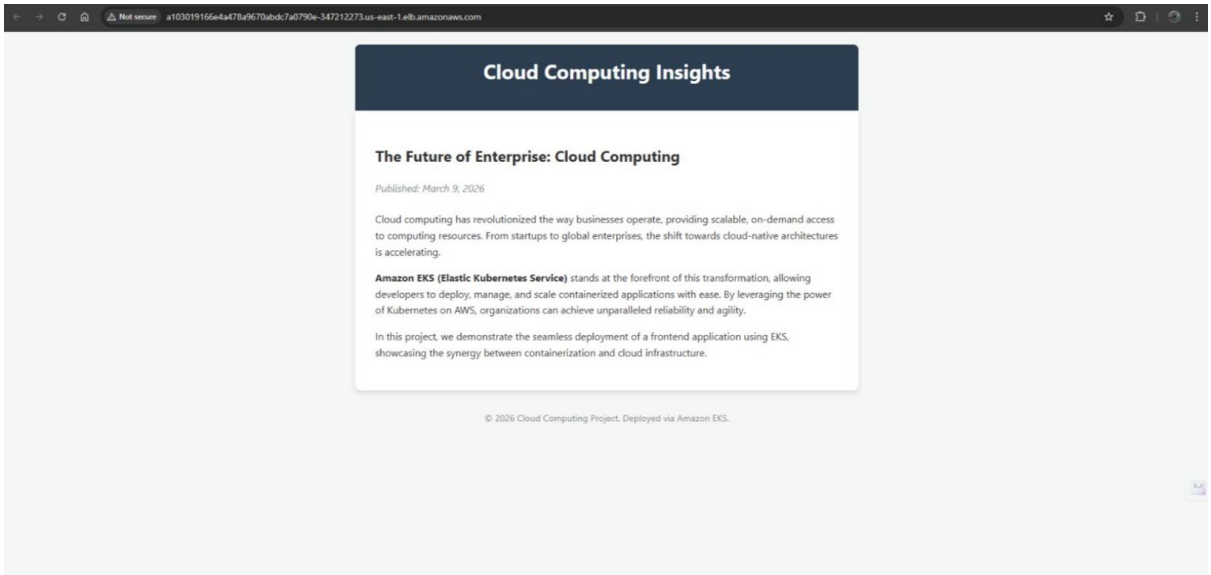
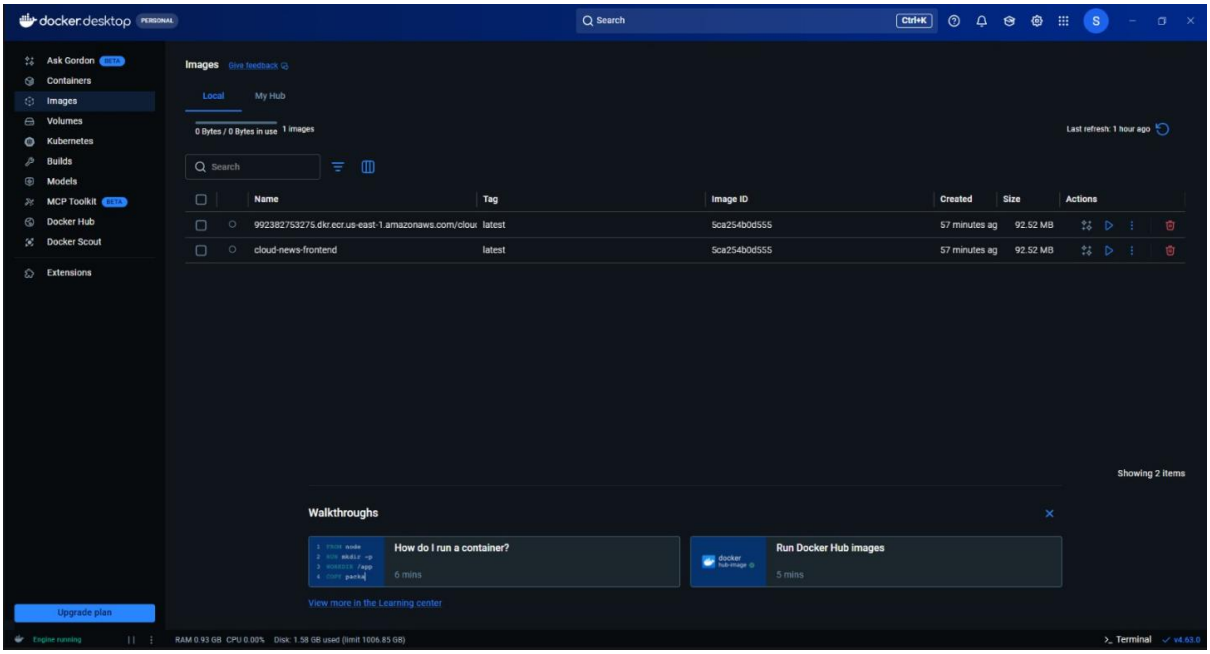
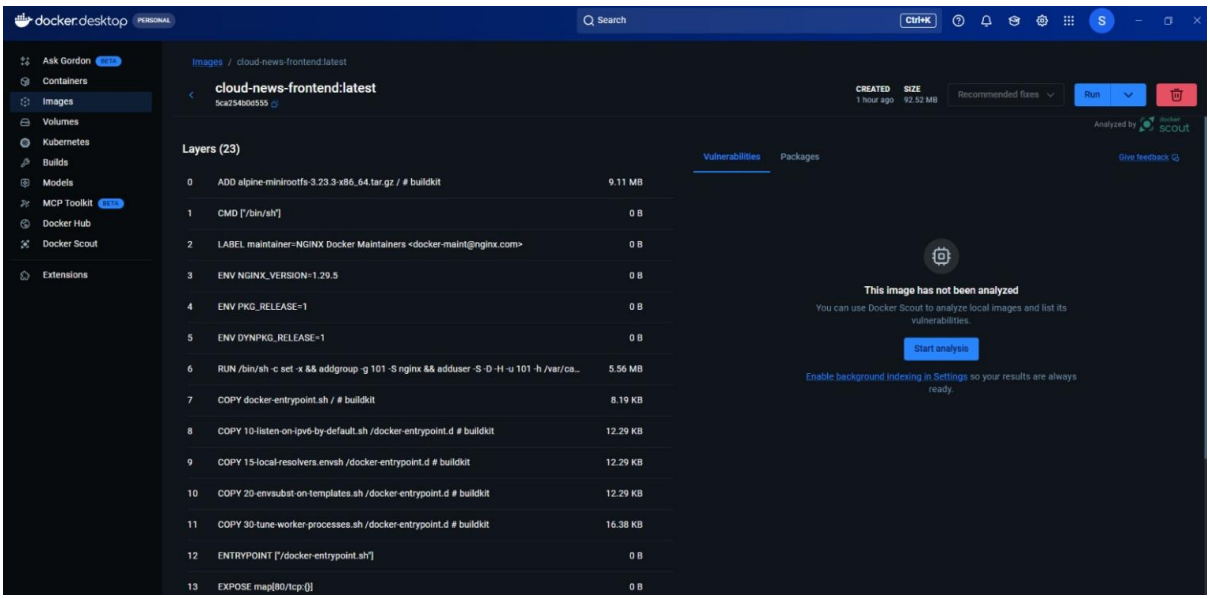
EC2 Instance :



Load Balancer :



Project Frontend :



Docker Desktop :